BRITISH STANDARD

BS EN 15267-1:2009

Air quality — Certification of automated measuring systems

Part 1: General principles

ICS 13.040.99

British Standards

National foreword

This British Standard is the UK implementation of EN 15267-1:2009.

The UK participation in its preparation was entrusted to Technical Committee EH/2/1, Stationary source emission.

A list of organizations represented on this committee can be obtained on request to its secretary.

This publication does not purport to include all the necessary provisions of a contract. Users are responsible for its correct application.

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Foreword

This document (EN 15267-1:2009) has been prepared by Technical Committee CEN/TC 264 “Air quality”, the secretariat of which is held by DIN.

This European Standard shall be given the status of a national standard, either by publication of an identical text or by endorsement, at the latest by September 2009, and conflicting national standards shall be withdrawn at the latest by September 2009.

Attention is drawn to the possibility that some of the elements of this document may be the subject of patent rights. CEN [and/or CENELEC] shall not be held responsible for identifying any or all such patent rights.

This document is Part 1 of a series of European Standards:

EN 15267-1, *Air quality — Certification of automated measuring systems — Part 1: General principles*

EN 15267-2, *Air quality — Certification of automated measuring systems — Part 2: Initial assessment of the AMS manufacturer’s quality management system and post certification surveillance for the manufacturing process*

EN 15267-3, *Air quality — Certification of automated measuring systems — Part 3: Performance criteria and test procedures for automated measuring systems for monitoring emissions from stationary sources*

According to the CEN/CENELEC Internal Regulations, the national standards organizations of the following countries are bound to implement this European Standard: Austria, Belgium, Bulgaria, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Norway, Poland, Portugal, Romania, Slovakia, Slovenia, Spain, Sweden, Switzerland and the United Kingdom.

Introduction

The certification of automated measuring systems (AMS) supports the requirements of certain Directives of the European Union (EU), which require, either directly or indirectly, that AMS comply with performance criteria, maximum permissible measurement uncertainties and testing requirements. These Directives include the Directive on the limitation of emissions of certain pollutants into the air from large combustion plants [1], the Directive on the incineration of waste [2] and the Framework Directive on ambient air quality assessment and management [3] and the associated daughter directives [4], [5], [6] and [7].

The responsibility for approving AMS for monitoring ambient air quality under Directive 96/62/EC lies with the national competent authority or a body designated by the EU member state. No explicit requirement for approving AMS for monitoring emissions from stationary sources is defined in the relevant EU Directives, although the competent authorities in some EU member states have such arrangements in place.

In some EU member states the competent authority delegates the responsibility for approval of AMS to a certification body accredited to EN 45011 by national accreditation bodies. In some EU member states the competent authority cannot be accredited by external bodies, in others they may be. These approaches have built up over many years and reflect the different administrative and legal arrangements that exist in the EU member states. In order to recognize these different approaches, this European Standard uses the collective term “relevant body” when referring to competent authority or certification body. The terms “competent authority” and “certification body” are only used where it is necessary to be specific for the purpose of clarity in the way in which a requirement applies under the different approaches.

The European Standard EN 45011 specifies general criteria that a certification body operating product certification shall follow if it is to be recognized at a national or European level as competent and reliable in the operation of a product certification system, irrespective of the sector involved. It is intended for the use of accreditation bodies concerned with recognizing the competence of certification bodies. EN 45011 is identical to ISO/IEC Guide 65. The document EA-6/01 [8] published by the International Accreditation Forum (IAF) provides guidelines on the application of EN 45011. The purpose of EA-6/01 is to harmonise the worldwide application of EN 45011 by accreditation bodies as an important step towards mutual recognition between certification bodies under the IAF Multilateral Agreement (MLA).

EN 45011 recognizes that these general criteria may have to be supplemented when applied to a particular sector. This European Standard provides guidance on the application of EN 45011 to the certification of AMS for monitoring ambient air quality and emissions from stationary sources. It is Part 1 of a three part series of European Standards, which specify common requirements for the certification of AMS in EU member states.

This European Standard defines common procedures and requirements for the certification of AMS to facilitate mutual recognition by the relevant bodies and thereby minimise administrative and cost burdens on AMS manufacturers seeking certification in multiple member states. It also describes the roles and responsibilities of manufacturers, test laboratories, certification bodies (for quality management systems) and relevant bodies under these procedures.

1 Scope

This European Standard specifies the general principles, including common procedures and requirements, for the product certification of automated measuring systems (AMS) for monitoring ambient air quality and emissions from stationary sources. This product certification consists of the following sequential stages:

a) performance testing of an automated measuring system;

b) initial assessment of the AMS manufacturer's quality management system;

c) certification;

d) surveillance.

This European Standard applies to the certification of all AMS for monitoring ambient air quality and emissions from stationary sources for which performance criteria and test procedures are available in European Standards.

2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

EN 14211, *Ambient air quality — Standard method for the measurement of the concentration of nitrogen dioxide and nitrogen monoxide by chemiluminescence*

EN 14212, *Ambient air quality — Standard method for the measurement of the concentration of sulphur dioxide by ultraviolet fluorescence*

EN 14625, *Ambient air quality — Standard method for the measurement of the concentration of ozone by ultraviolet photometry*

EN 14626, *Ambient air quality — Standard method for the measurement of the concentration of carbon monoxide by nondispersive infrared spectroscopy*

EN 14662-3, *Ambient air quality — Standard method for measurement of benzene concentrations — Part 3: Automated pumped sampling with in situ gas chromatography*

EN 15267-2, *Air quality — Certification of automated measuring systems — Part 2: Initial assessment of the AMS manufacturer's quality management system and post certification surveillance for the manufacturing process*

EN 15267-3, *Air quality — Certification of automated measuring systems — Part 3: Performance criteria and test procedures for automated measuring systems for monitoring emissions from stationary sources*

EN 45011, *General requirements for bodies operating product certification systems (ISO/IEC Guide 65:1996)*

EN ISO 9001, *Quality management systems — Requirements (ISO 9001:2000)*

EN ISO/IEC 17021, *Conformity assessment — Requirements for bodies providing audit and certification of management systems (ISO/IEC 17021:2006)*

EN ISO/IEC 17025, *General requirements for the competence of testing and calibration laboratories (ISO/IEC 17025:2005)*

3 Terms and definitions

For the purposes of this document, the following terms and definitions apply.

3.1
automated measuring system
AMS
entirety of all measuring instruments and additional devices for obtaining a result of measurement

NOTE 1 Apart from the actual measuring device (the analyser), an AMS includes facilities for taking samples (e.g. probe, sample gas lines, flow meters and regulator, delivery pump) and for sample conditioning (e.g. dust filter, pre-separator for interferents, cooler, converter). This definition also includes testing and adjusting devices that are required for functional checks and, if applicable, for commissioning.

NOTE 2 The term “automated measuring system” (AMS) is typically used in Europe. The terms “continuous emission monitoring system” (CEM) and “continuous ambient-air-quality monitoring system” (CAM) are also typically used in the UK and USA.

3.2
relevant body
competent authority or certification body, nominated by a competent authority or EU member state, that carries out the certification of automated measuring systems

3.3
competent authority
organisation which implements the requirements of EU Directives and regulates installations, which must comply with the requirements of applicable European Standards

3.4
certification body
any body operating a product certification system or any body accredited to EN ISO/IEC 17021 for the certification of quality management systems

3.5
manufacturer
organisation, situated at a stated location or locations, that carries out or controls such stages in the manufacture, assessment, handling and storage of a product that enables it to accept responsibility for continued compliance of the product and its certification, and undertakes all obligations in that connection

NOTE The term "manufacturer" is used instead of "organisation" as used in EN ISO 9001. For the purpose of this document they are interchangeable.

3.6
test laboratory
laboratory accredited to EN ISO/IEC 17025 for carrying out the performance tests on automated measuring systems in accordance with applicable European Standards

3.7
product
automated measuring system

3.8
technical file
record of the reference documents and design changes to the reference documents

3.9
reference document
document that controls the manufacture and design of an AMS and is referenced in the test report

NOTE Reference documents can include drawings, specifications, instructions and computer code.

3.10
related document
document not referenced in the test report

NOTE A related document can be used, for example, for the detailed manufacture of component parts.

3.11
certification range
range over which the automated measuring system is tested and certified for compliance with the relevant performance criteria

NOTE 1 The lower limit is typically the detection limit of the AMS and often considered to be zero.

NOTE 2 Generally, the lower the certification range, the better the performance of the AMS. Also an AMS typically performs satisfactorily at higher values over the measurement range.

3.12
surveillance
systematic iteration of conformity assessment activities as a basis for maintaining the validity of the statement of conformity

[EN ISO/IEC 17000:2004, 6.1]

NOTE For the purposes of this European Standard surveillance focuses on the manufacturer's quality management system to ensure that automated measuring systems continue to comply with the standard to which they are certified.

3.13
legislation
directives, acts, ordinances and regulations

4 Abbreviations

AMS automated measuring system

EA European Co-operation for Accreditation

EU European Union

IAF International Accreditation Forum

MLA Multilateral Agreement

QMS quality management system

5 Principles

5.1 Performance testing of the automated measuring system

Performance testing consists of a combination of laboratory and field testing. Laboratory testing is designed to assess whether an automated measuring system (AMS) can meet, under controlled conditions, the performance criteria specified for the relevant performance characteristics. Field testing, over a minimum three month period, is designed to assess whether an AMS can continue to work and meet the relevant performance criteria in a real application. For emissions monitoring, AMS field testing is carried out on an industrial installation representative of the intended application for the AMS for which the manufacturer seeks certification. For ambient air monitoring AMS field testing is carried out at sites as specified in the Framework Directive on ambient air quality assessment and management [3] and the associated daughter directives [4], [5], [6] and [7] as well as in the corresponding European Standards.

An AMS is tested to evaluate its performance against criteria specified in an applicable European Standard for the performance and testing of AMS. The testing is carried out by a test laboratory accredited to EN ISO/IEC 17025 with test procedures specified in the applicable Standard.

The applicable standards include:

— EN 15267-3 for AMS for monitoring emissions from stationary sources,

— EN 14211, EN 14212, EN 14625, EN 14626 and EN 14662-3 for AMS for monitoring ambient air quality, and

— other European Standards currently under preparation that specify performance criteria and test procedures for AMS for monitoring emissions from stationary sources or monitoring ambient air quality.

5.2 Initial assessment of the AMS manufacturer's quality management system

An AMS typically undergoes design changes during its production life and it is essential to ensure that after such changes, the AMS still meets the required performance criteria. Therefore, in order to meet these performance criteria the manufacturer controls the production and design of the AMS using a quality management system, which meets the requirements specified in EN 15267-2.

Compliance with the requirements of EN 15267-2 is determined by an initial assessment of the manufacturer’s quality management system. This initial assessment can be carried out by the relevant body, or a party acting on behalf of the relevant body, or by a certification body accredited to EN ISO/IEC 17021 if the requirements of EN 15267-2 have been incorporated into a certified quality management system (e.g. EN ISO 9001). In the latter case, the initial assessment can focus only on the additional requirements of EN 15267-2.

5.3 Certification

The reports from the testing of the AMS and the initial assessment of the manufacturer’s quality management system are reviewed by the relevant body for conformance with the applicable standards. If satisfactory, then the relevant body issues a certificate of conformance for the AMS.

If manufacturers, test laboratories, and relevant bodies meet the requirements of the EN 15267 series of standards, then such compliance provides the means for mutual recognition between relevant bodies in EU member states.

The relevant body should deal with applications based on mutual recognition in ways that minimise the costs and administrative burden on the manufacturer. These should involve using simple administrative procedures to verify that all necessary details, certificates and reports have been submitted.

The relevant bodies in EU member states are recommended to establish a central register listing certified AMS. The register should provide traceable links to the original certificates and supporting information.

5.4 Surveillance

Following certification, the relevant body ensures that surveillance of the manufacturing process and performance of certified AMS is carried out periodically.

This post-certification surveillance is normally carried out by the organisation which performed the initial assessment of the manufacturer’s quality management system to EN 15267-2 as described in 5.2.

6 Roles and responsibilities

6.1 General

The roles and responsibilities of the AMS manufacturer, the test laboratory and the relevant body are described below. The manufacturer shall be independent of the relevant body, and any party acting on behalf of the relevant body, and of the test laboratory. The roles and responsibilities of the relevant body and test laboratory can be performed by a single organisation if it is suitably accredited.

6.2 Roles and responsibilities of the manufacturer

The manufacturer shall

— make the initial approach to a test laboratory and/or relevant body for performance testing of an AMS,

NOTE 1 Whether a test laboratory or relevant body is approached first it is advisable that, with the agreement of the manufacturer, they inform the other so that the scope of the testing and the intended application (i.e. ambient air monitoring, type of industrial installation) for which certification is to be sought can be decided in consultation. When the scope and application have been agreed, testing can proceed.

— submit two identical AMS for performance testing to a test laboratory and provide all necessary information for testing,

NOTE 2 The manufacturer can submit more than two identical AMS, to complete the testing faster by running the laboratory and field testing in parallel.

— establish, maintain and operate a quality management system meeting the requirements of EN 15267-2,

— obtain an initial assessment of the quality management system to demonstrate its compliance with the requirements of EN 15267-2 by the relevant body, or a party acting on its behalf, or by a certification body accredited to EN ISO/IEC 17021 if the requirements of EN 15267-2 have been incorporated into a certified quality management system,

— ensure quality assurance and control of manufacturing such that all certified AMS continue to meet the applicable performance criteria,

— control and assess design changes, and keep detailed records of changes to the AMS within a technical file for each certified AMS in accordance with the requirements of EN 15267-2,

NOTE 3 The assessment can include complete or partial re-testing of the changed AMS. The manufacturer can seek the advice of a test laboratory and the relevant body regarding the need for complete or partial re-testing, following design changes. Any re-testing can be carried out by the manufacturer or, at the manufacturer’s request, by a test laboratory.

— record the methods and results of re-testing in the technical file, as specified in EN 15267-2, if an AMS requires partial or complete re-testing, and

— notify a test laboratory (preferably the one that carried out the initial tests for certification) and the relevant body of changes to the AMS.

6.3 Roles and responsibilities of the test laboratory

The test laboratory shall

— establish, maintain and operate a management system accredited to EN ISO/IEC 17025, where the scope of accreditation includes the applicable standards for monitoring and testing, NOTE 1 For the testing of AMS for monitoring emissions from stationary sources, the applicable standards for monitoring and testing include EN 15267-3, EN 14884, and other European Standards, currently under preparation, that specify performance criteria and test procedures for AMS for monitoring other emissions from stationary sources, and CEN/TS 15675, which elaborates EN ISO/IEC 17025 for application to stack emission measurements, which form part of the field testing of AMS.

NOTE 2 For the testing of AMS for monitoring ambient-air quality, the applicable standards for monitoring and testing include EN 14212, EN 14211, EN 14626, EN 14625 and EN 14662-3 and other European Standards, currently under preparation, that specify performance criteria and test procedures for AMS for monitoring ambient air quality.

— evaluate AMS conformity with the performance criteria defined in applicable European Standards, by testing in accordance with the test procedures specified in those standards,

— guide the manufacturer on the suitability of the AMS for different applications (i.e. ambient air monitoring, industrial installations) and measurement ranges in accordance with the requirements defined in the applicable standard,

— make applications to the relevant body for the certification of AMS, if required and on behalf of the manufacturer, and

— evaluate any design changes to the AMS, if requested by the manufacturer, and, if they affect the original certification, notify the relevant body of such changes and advise the manufacturer and relevant body if any re-testing is required.

6.4 Roles and responsibilities of the relevant body

The relevant body shall

— be accredited to EN 45011 for the certification of AMS in accordance with the requirements of this standard, if the relevant body is a certification body,

— have in place appropriate procedures for the certification of AMS in accordance with the requirements of this standard, if the relevant body is an unaccredited competent authority,

— provide guidance on the arrangements and requirements for certification to the manufacturer, test laboratory and the certification body for quality management systems if the manufacturer has incorporated the requirements of EN 15267-2 into a certified quality management system,

— assess test reports and determine whether the test laboratory is appropriately accredited to carry out the tests and whether the data support the application for certification including the proposed scope,

— verify the evidence that the AMS manufacturer has a quality management system which complies with the requirements of EN 15267-2,

NOTE 1 If a manufacturer has incorporated the requirements of EN 15267-2 into a certified quality management system (e.g. EN ISO 9001) the initial assessment of its quality management system can be carried out by a third-party certification body accredited to EN ISO/IEC 17021.

NOTE 2 If a manufacturer already has a certified quality management system, the initial assessment can focus only on the additional requirements of EN 15267-2.

— liaise as appropriate with the relevant national competent authority,

— issue certificates with an appropriate scope of certification,

NOTE 3 The scope of certification includes the measured components, certification ranges, process applications and any limitations of use.

— issue certificates in at least one of the three principal CEN languages, i.e. English, French or German,

— add the AMS to its official register of certified AMS, and

— ensure that post-certification surveillance of the manufacturing process and performance of certified AMS is periodically carried out.

7 Certification procedure

7.1 General

Figure A.1 illustrates the main stages involved in obtaining certification of an automated measuring system.

7.2 Performance testing of the automated measuring system

At the time of application for testing, the manufacturer shall agree with the test laboratory and the relevant body the ranges and applications of the AMS, according to the requirements of European Standards, legislation and the national competent authority, for which certification will be sought.

NOTE The manufacturer can approach the test laboratory first and request the laboratory to act on its behalf with the relevant body or vice versa. Whichever route is chosen it is important that all parties are kept fully informed.

The testing shall be carried out according to the requirements of the applicable European Standard. If an AMS fails any of the tests, then the testing shall be halted. If the manufacturer addresses the reasons for any failures and resubmits the AMS, the testing shall be started again from a point in the programme agreed to be appropriate by the test laboratory and the relevant body in consultation with the manufacturer. Details of this re-start shall be recorded in the test report.

Upon completion of the test programme, the test laboratory shall write a report of the testing and issue this to the manufacturer and the relevant body. If the AMS meets the requirements of the applicable European Standard, then the test report shall include a recommendation for certification, stating the applicable ranges (i.e. certification and supplementary ranges), the suitability of the AMS for specified fields of application, and any qualifying remarks.

7.3 Initial assessment of the manufacturer's quality management system

The manufacturer shall have its quality management system assessed for compliance with EN 15267-2.

If the requirements of EN 15267-2 are incorporated into an uncertified quality management system the manufacturer should request the relevant body to arrange the initial assessment.

If the requirements of EN 15267-2 are incorporated into a certified quality management system (e.g. EN ISO 9001) the manufacturer can request a third party certification body accredited to EN ISO/IEC 17021 to carry out the initial assessment. In this case the initial assessment should focus only on the additional requirements of EN 15267-2. If this route is chosen it is important that the relevant body is kept informed.

If a third party assessment is obtained the manufacturer shall make the assessment report available to the relevant body as part of its final application for certification.

7.4 Certification

The relevant body shall review the application, the test report, the assessment of the manufacturer's quality management system and any other supporting documents.

The information to be supplied by the manufacturer when submitting an application shall include the following:

— the intended certification range and any supplementary ranges;

— the measured components and industrial process applications or ambient air monitoring for which certification is being sought;

— a description of the AMS;

— the results of the performance testing of the AMS, as required by the applicable standard;

— evidence of compliance of the quality management system to EN 15267-2;

— the manufacturer's documents and related documents.

NOTE Manufacturer's documents can be instructions, data sheets and sales literature.

If there is sufficient evidence to certify the AMS, then the relevant body shall issue a certificate of conformity to the manufacturer and add the AMS to its official register of certified AMS.

An example of a certificate template is given in Annex B.

The certificate shall clearly state the following:

— a unique identifier for the type of the automated measuring system;

— the European Standard specifying the performance criteria;

— the measurement ranges, minimum certification ranges (and supplementary ranges if applicable) for each certified measured component;

— the applicable industrial processes or ambient air monitoring applications;

— the suitability of automated measuring systems tested in accordance with EN 15267-3 for installations under EU Directives for large combustion plant and/or waste incineration;

— a specific reference to the performance test report from the test laboratory.

7.5 Surveillance

Following certification the relevant body shall ensure that surveillance of the manufacturing process and performance of the certified AMS is periodically carried out to determine continued conformance with the applicable European Standard. This surveillance shall involve audits of the manufacturer's quality control of design changes, manufacture of the AMS, and effective responses to complaints from users.

Surveillance shall be carried out at least every year. Annual audits shall be carried out at the manufacturer's premises (where the AMS is manufactured and tested; see 3.5) in accordance with the requirements of EN ISO 19011. A detailed examination of the conformity of the manufactured AMS with the applicable performance criteria shall be carried out. The details of any discrepancies capable of affecting the performance of the AMS with respect to the applicable performance criteria shall be noted. In the event that non-conformance with the applicable performance criteria is found then the manufacturer shall be required to carry out any corrective actions. These actions shall be formally closed out by further audit. In the event that a further audit shows that the non-conformances have not been closed out the relevant body shall consider revoking the AMS's certificate.

NOTE During an audit the auditor can witness the effectiveness of tests carried out by the manufacturer on a certified AMS.

7.6 Review of certifications

The relevant body shall keep the validity of the certification of the AMS under continuous review taking into account the reports from technical changes to the AMS, post-certification surveillance, any changes in the technical requirements notified by competent authorities, and complaints from users.

Annex A
(informative)

Application and assessment processes

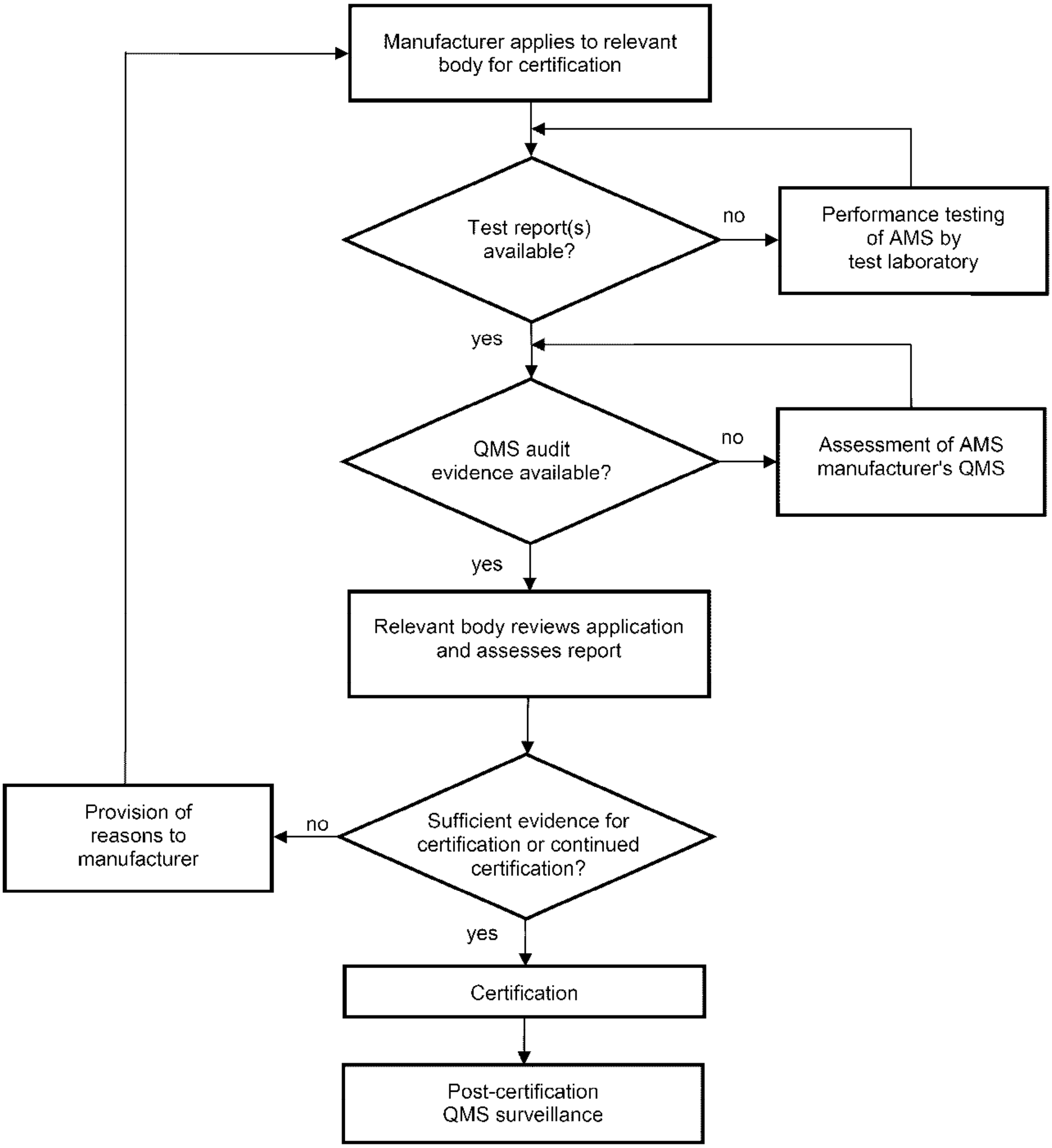

Figure A.1 — Illustration of the application and assessment processes

Annex B
(informative)

Example certificate template

PRODUCT CONFORMITY CERTIFICATE

This is certifying that the AMS:

[*AMS name*]

manufactured by:

[*manufacturer's name and address*]

has been assessed by [*relevant body's name*] and found to comply with:

[*applicable European Standard with publication date*]

Certification Ranges:

[certified ranges]

Certification is awarded in respect of the conditions stated in this certificate.

Project No:
Certificate No:
Initial Certification:
This Certificate Issued:
Certifying officer: [*signature applied here*]
[*name of certifying officer*]

[*name, address, telephone and fax numbers and email address as appropriate for the relevant body*]

Approved application

The application for certification sought by the manufacturer was for use on a [*installation/process type or ambient air application*].

The suitability of the product for this application was assessed on the basis of a three months test of the [*AMS name*] on [*type of installation/process or ambient air application*].

Any potential user should ensure, in consultation with the manufacturer that this AMS is suitable for the [*installation/process or ambient air application*] on which it will be installed.

Basis of Certification

This certification is based on the following test report(s) and on [relevant body's name] assessment and ongoing surveillance of the product and the manufacturing process:

Test report: [*test laboratory's name and address*]

Report No: [*unique test report number and date of issue*]

Certified product

This certificate applies to AMS conforming to the following description:

[*general description of certified AMS*]

General notes

This certificate is based upon the equipment tested. The manufacturer is responsible for ensuring that on-going production complies with the European Standard defined in this certificate. The manufacturer is required to maintain an approved quality management system controlling the manufacture of the certified product. Both the product and the quality management systems shall be subject to regular surveillance.

If a certified product is found no longer to comply with the applicable European Standard, [*relevant body's name*] should be notified immediately at the address shown on page 1.

The certification marks that can be applied to the product or used in publicity material are defined in [*relevant body's name*] regulations.

This document remains the property of [*relevant body's name*] and shall be returned when requested by the relevant body.

Bibliography

[1] Directive 2001/80/EC of the European Parliament and of the Council of 23 October 2001 on the limitation of emissions of certain pollutants into the air from large combustion plants, OJ L 309, p. 1

[2] Directive 2000/76/EC of the European Parliament and of the Council of 4 December 2000 on the incineration of waste, OJ L 332, p. 91

[3] Council Directive 96/62/EC of 27 September 1996 on ambient air quality assessment and management, OJ. L 296, p. 55

[4] Council Directive 1999/30/EC of 22 April 1999 relating to limit values for sulphur dioxide, nitrogen dioxide and oxides of nitrogen, particulate matter and lead in ambient air, OJ L 163, p. 41

[5] Directive 2000/69/EC of the European Parliament and of the Council of 16 November 2000 relating to limit values for benzene and carbon monoxide in ambient air, OJ L 313, p. 12

[6] Directive 2002/3/EC of the European Parliament and of the Council of 12 February 2002 relating to ozone in ambient air, OJ L 67, p. 14

[7] Directive 2004/107/EC of the European Parliament and the Council of 15 December 2004 relating to arsenic, cadmium, mercury, nickel and polycyclic aromatic hydrocarbons in ambient air, OJ L 23, p. 3

[8] EA-6/01:1999, *EA Guidelines on the application of EN 45011*

[9] EN ISO/IEC 17000:2004, *Conformity assessment — Vocabulary and general principles (ISO/IEC 17000:2004)*

[10] EN 14884, *Air quality — Stationary source emissions — Determination of total mercury: automated measuring systems*

[11] CEN/TS 15675, *Air quality — Measurements of stationary source emissions — Application of EN ISO/IEC 17025:2005 to periodic measurements*

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